Magneto-statics

Dr. Sampath Kumar

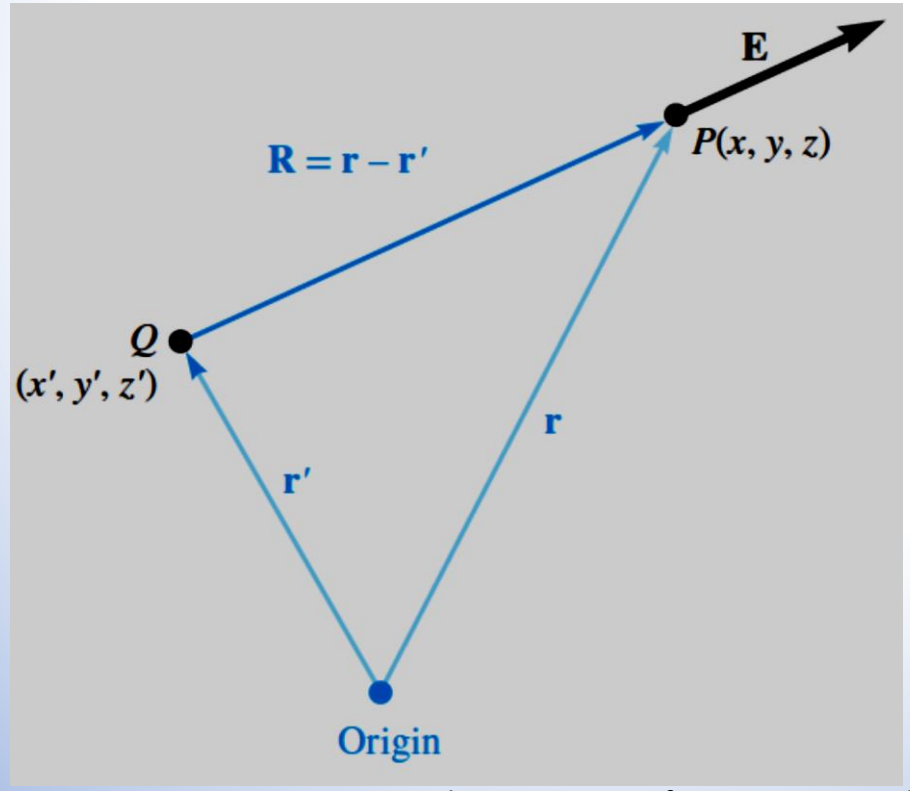
Associate Professor

Department of ECE

MIT, Manipal

Electrostatics – Magneto-statics

- Static fields
- Time varying fields – constitutes waves
- Recap



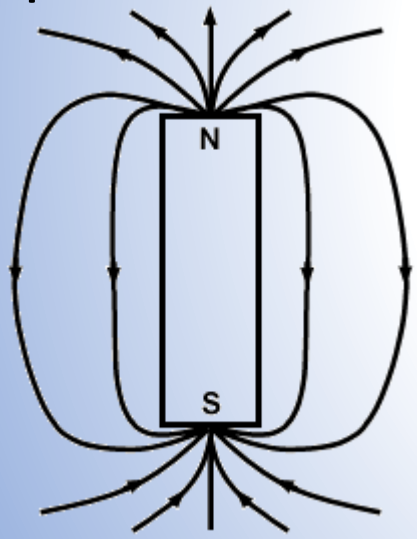
$$\mathbf{E} = \frac{Q_1}{4\pi\epsilon_0 r^2}\mathbf{a}_r$$

The Steady Magnetic Field

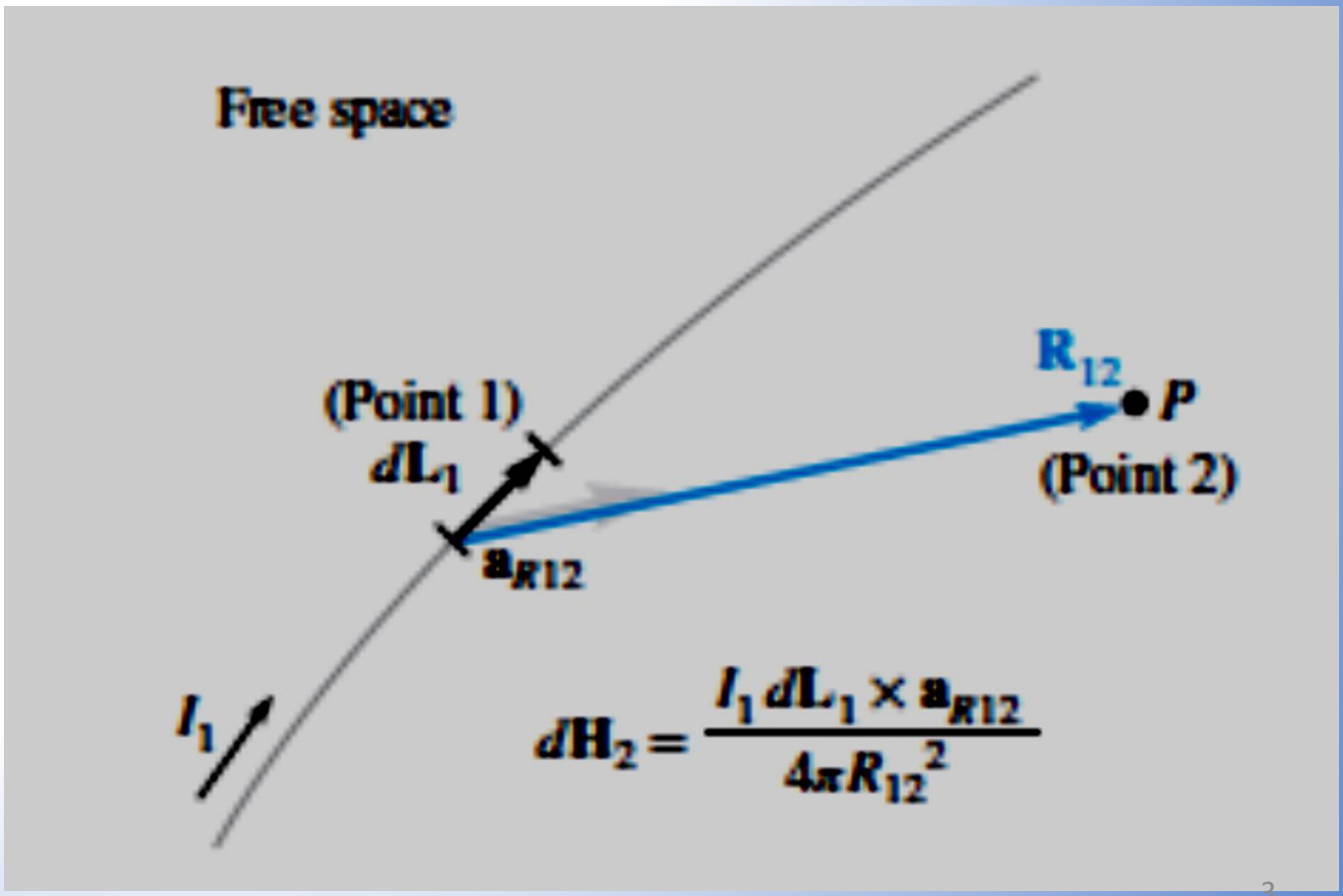
BIOT-SAVART LAW

$$\mathbf{E} = \frac{Q_1}{4\pi\epsilon_0 r^2}\mathbf{a}_r$$

- Source of steady magnetic field – permanent magnet



- Magnetic filed also produced by differential DC element in free space

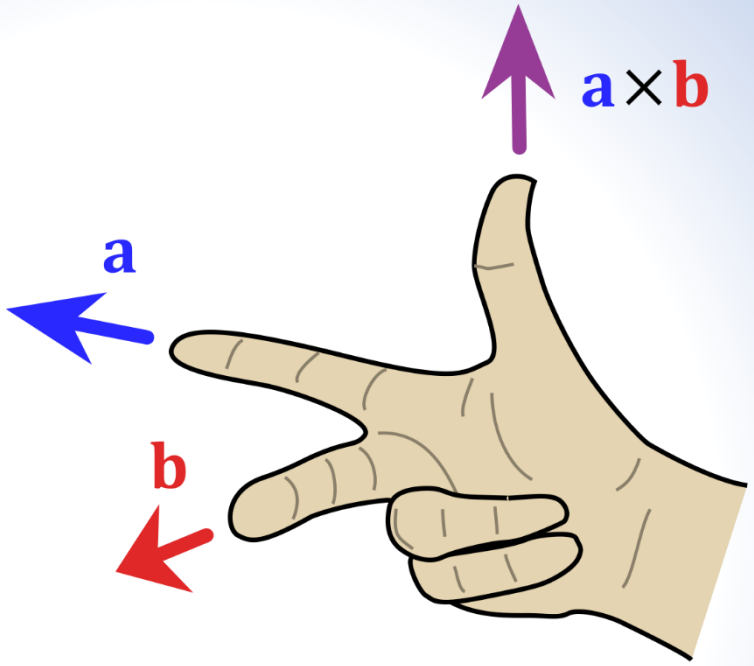


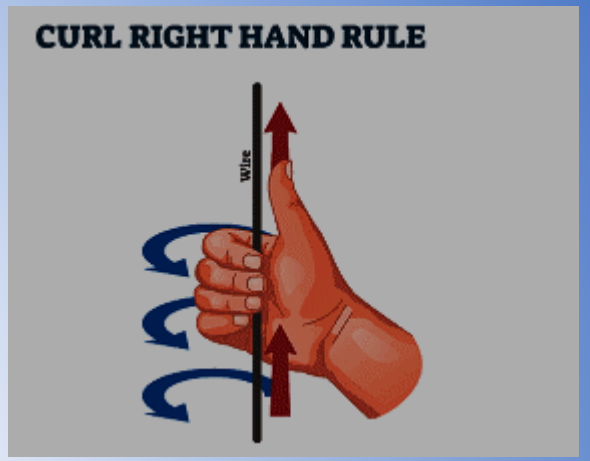
Biot-Savart Law

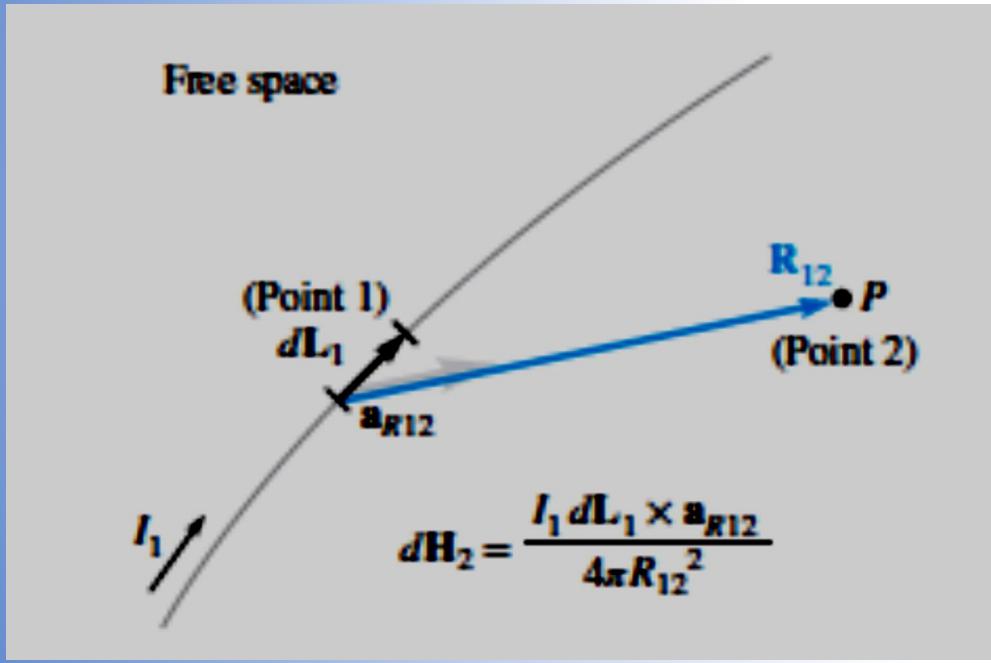
- At any point *P* the magnitude of the magnetic field intensity produced by the differential element is proportional to the product of the current, the magnitude of the differential length, and the sine of the angle lying between the filament and a line connecting the filament to the point *P* at which the field is desired;
- Also, the magnitude of the magnetic field intensity is inversely proportional to the square of the distance from the differential element to the point *P*.
- The direction of the magnetic field intensity is - direction of progress of a right-handed screw turned from $d\mathbf{L}$ through the smaller angle to the line from the filament to *P*.

$$d\mathbf{H} = \frac{Id\mathbf{L} \times \mathbf{a}_R}{4\pi R^2} = \frac{Id\mathbf{L} \times \mathbf{R}}{4\pi R^3}$$

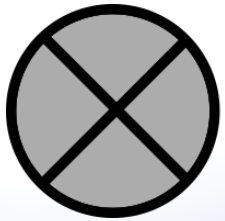
Direction of cross product





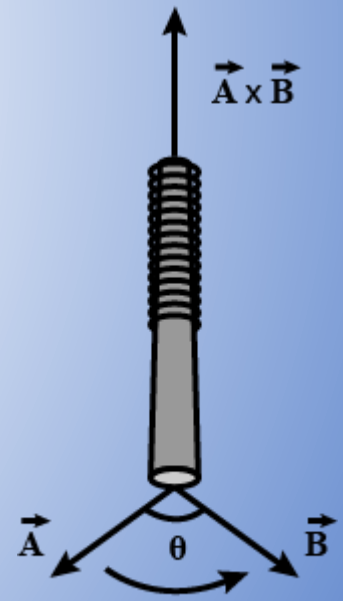


Going in to the paper





Coming out of paper



Vector Product

In the above case H field is going in to the paper

- Magnetic filed intensity H → unit A/M (E→ V/m)
- WKT Current continuity equation = $\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$
- Differential current elements can't be isolated (Current flows from higher potential to lower potential) → $\nabla \cdot \mathbf{J} = 0$
- Using divergence theorem

$$\oint_s \mathbf{J} \cdot d\mathbf{S} = 0$$

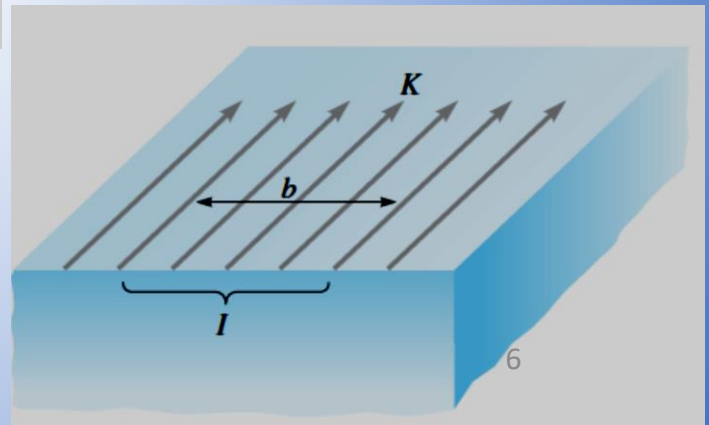
$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{A}\, dv$$

- Also, The total current flowing in a circuit gives,

Integral form of Biot-Savart Law =

$$\mathbf{H} = \oint \frac{Id\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

- Define surface current density K → K=I/b
- IdL = KdS = JdV



$$\mathbf{H} = \oint \frac{Id\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

$$\mathbf{H} = \int_s \frac{\mathbf{K} \times \mathbf{a}_R dS}{4\pi R^2}$$

$$\mathbf{H} = \int_{\text{vol}} \frac{\mathbf{J} \times \mathbf{a}_R dv}{4\pi R^2}$$


Thank
you